PalmSource™ 99

Latest Palm OS™ SW: Localizing Applications

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What's This Talk About?

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- Showing how easy it is to expand into the international market
- Introducing new overlay manager and tools

Topics Covered

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- The overlay concept
- Overlay resource databases
- How to develop overlays
- What overlaid software looks like to the user
- Localization challenges
- Whatever I forget to say (Q & A)

The Overlay Concept

- Wouldn’t it be great if you didn’t have to rebuild an application to localize it?
- What if you could just install a separate resource database containing the localized resources?
- That’s it? My distributor could probably handle that for me!

Overlay Resource Databases

- Separate resource databases containing localized resources
- One-to-one mapping from base resource DB to overlay resource DB
  - Any app, panel, extension or library can be overlaid
  - Overlay DB name includes “locale suffix” (e.g. “Address Book_enUS”)
- Overlay manager automatically substitutes localized resources at runtime
  - No need to ever call overlay manager
  - If overlay resource DB exists, localized resources are returned by data manager

Overlay Specification Resources

- Establish link between base and overlay DBs
  - Required for overlay DB
  - Optional for base DB
- Type = ‘ovly’ ID = 1000
- Contents:
  - Target “locale” (language and country)
  - Info about base DB (type, creator, checksum, etc.)
  - Info about each overlaid resource
    - Normally they specify replacement, but resources can also be added
    - Base’s ‘ovly’ specifies set of “localizable” resources

Developing Overlays

- Constructor support is coming soon to a desktop near you
- Use PRC2OVL desktop application for now
  - Create (unlocalized) overlay .prc with PRC2OVL
  - Localize the .prc (e.g. with OnBoard RsrcEdit:)
    - Install overlay DB in POSE
    - Localize overlay DB with OnBoard RsrcEdit
    - Export to desktop (e.g. with POSE DB Export)

Other Ways to Build Overlays

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- Overlayer handheld application
  - Create (unlocalized) overlay DB with Overlayer running in Palm OS™ Emulator
  - Localize overlay DB with OnBoard RsrcEdit…
- MakeOverlay desktop application
  - Obtain base resource database (.prc file)
  - Obtain and localize input resources (.rsrc files)
  - Drop folder containing above onto MakeOverlay
- MakeOverlay MPW tool

Other Useful Tools

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- PrcChecker desktop application
  - Verifies format of .prc files
  - Displays database header info
  - Displays list of enclosed resources
  - Displays contents of ‘ovly’ (1000) resource
- PrcReversi desktop application
  - Creates enUS overlay with each text word reversed
  - Useful for quickly finding potential localization problems

What Overlaid Software Looks Like

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- Your application appears localized when installed on a device with target locale ROM
  - Even its name, icon, and default data (tAIN, tAIB, and dflt resources)
- Overlay DBs are automatically deleted along with base DBs (e.g., applications)
  - Overlay DBs aren't visible in Launcher
- Overlay DBs cannot be beamed
- Future: multi-lingual ROMs?

Localization Challenges

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- Hard-coded text (and other data) resources
- Use text manager to parameterize messages
- Leave room in forms for text expansion
- Track changes to base resources carefully
- Come to “Designing for Localization” session

Summary

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- Localization is no longer dependent on the development team
  - Localizations can be performed without a development environment
  - Localizations can even be performed by distributor without modifying your base application
- The development team should maintain this independence
  - Keep localizable data in resources
  - Track changes to those resources


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